❶ Agent 1's verified session

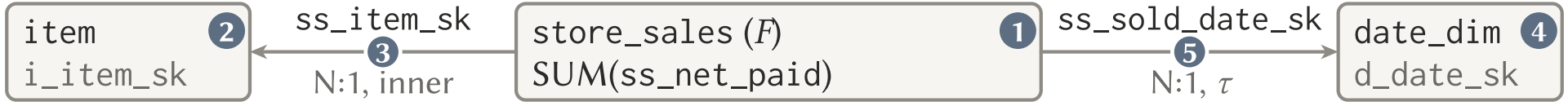
answer 2,754,299.60

```
SELECT SUM(ss_net_paid) FROM store_sales JOIN item ... JOIN date_dim ...
WHERE i_category = 'Electronics'  φ  AND d_year = 2001 AND d_moy = 1  p
```

G3 keeps φ in I; filters on date_dim and the question's literals become p

Implementation I as a join tree

c_i: derived conditions



❷ Conditions $C(m^r)$ measured at publication

kept in H at (id(c), V(dep(c)))

❶ KeyUnique(store_sales.K)
grain: 1,000,000 rows, 1,000,000 keys

❸ Unmatched(store_sales ⟶ item)
join completeness: 0 of 10^6: $b = 0.00\%$

❺ Unmatched(store_sales ⟶ date_dim)
date coverage: 10,305 of 10^6: $b = 1.03\%$

❷ KeyUnique(item.i_item_sk)
join cardinality: 1,000 rows, 1,000 keys

❹ KeyUnique(date_dim.d_date_sk)
date mapping: 1,096 rows, 1,096 keys

accept KeyUnique: rows = keys;
Unmatched: loss $\le b + 0.1$ pt

G1 basis · G2 chain · G3 verified joins, φ vs. p, baselines b · G4 = c_1 · G5 read-only
G6 session SQL, G7 Compile(I, p_0) ⟶ 2,754,299.60 ✓ draft without φ ⟶ 13,532,799.00 ✗